

DSA 2204: ARTIFICIAL INTELLIGENCE

Lecturer's Details

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Course Learning Outcomes

1. Explain Artificial Intelligence concept and its applications.
2. Represent knowledge using various techniques.
3. Apply Artificial Intelligent techniques in solving problems of a particular domain.
4. Analyze business problems and determine which Artificial Intelligence techniques are appropriate to solve them.

Assessment Mode

Continuous Assessment Tests		50%
CATs (2)	30%	
Assignments (3)	20%	
Final Examination		50%
Total		100%

What is Artificial Intelligence?

- ❖ Making computers that think?
- ❖ The automation of activities we associate with human thinking, like decision making, learning ... ?
- ❖ The art of creating machines that perform functions that require intelligence when performed by people ?
- ❖ The study of mental faculties using computational models?
- ❖ The study of computations that make it possible to perceive, reason and act?
- ❖ A field of study that seeks to explain and emulate intelligent behaviour in terms of computational processes?

What is Artificial Intelligence?

- ❖ AI is based on the Four Classical Definitions of AI, proposed by Stuart Russell and Peter Norvig in *Artificial Intelligence: A Modern Approach*.
- ❖ These four definitions form a 2×2 matrix along two dimensions:
 - ❖ **Thought vs. Behavior** — internal reasoning processes vs. external observable actions
 - ❖ **Human vs. Rational** — measured by human standards vs. measured by an ideal of rationality

What is Artificial Intelligence?



THOUGHT

How the system reasons internally



Systems that think like humans

Systems that model human cognitive processes, such as perception, memory, and reasoning.

Examples: Cognitive architectures, ACT-R, Theory of Mind models



Systems that think rationally

Systems that use formal logic and optimal reasoning to reach correct conclusions.

Examples: Theorem provers, logic-based systems, planners



BEHAVIOUR

How the system behaves outwardly



Systems that act like humans

Systems that emulate human behaviour, often indistinguishable from a human observer.

Examples: Chatbots, virtual assistants, Turing Test systems



Systems that act rationally

Systems that behave to achieve goals in the best possible way using available information.

Examples: Autonomous agents, optimisation systems, robots

HUMAN

Inspired by or emulating human characteristics

RATIONAL

Based on logic, rules, and optimality

What is Artificial Intelligence?

Systems that act like humans - Turing Test:

- ❖ **Goal:** Make machines behave in ways indistinguishable from humans.
- ❖ Proposed by **Alan Turing (1950)**. The Turing Test is a method of inquiry in AI for determining whether a computer can think like a human being.
- ❖ Turing proposed that a computer can be said to possess artificial intelligence if it can mimic human responses under specific conditions.
- ❖ **Example:** The CAPTCHA system - commonly used online to tell humans and bots apart on the internet.

What is Artificial Intelligence?

- ❖ The original Turing Test required three terminals, each of which was physically separated from the other two.
- ❖ One terminal was operated by a computer, while the other two were operated by humans.
- ❖ During the test, one of the human's functions as the questioner, while the second human and the computer functions as respondents.
- ❖ The questioner interrogates the respondents within a specific subject area, using a specified format and context.
- ❖ After a preset length of time or number of questions, the questioner is then asked to decide which respondent was human and which was a computer.

What is Artificial Intelligence?

- ❖ The test is repeated many times with different judges. If the questioner cannot reliably tell the machine from the human (Turing originally proposed about 30% of judges being fooled within 5 minutes), the computer is considered to have demonstrated artificial intelligence.
- ❖ The capabilities the test was required to pass:
 - ❖ **Natural Language Processing** - to communicate.
 - ❖ **Knowledge Representation** - to store information.
 - ❖ **Automated Reasoning** - to answer questions using stored knowledge.
 - ❖ **Machine Learning** - to adapt to new circumstances

What is Artificial Intelligence?

Systems
that act
like
humans:

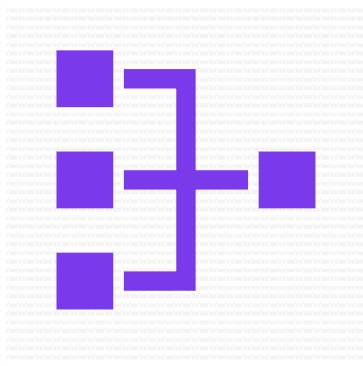
The Turing Test Approach

- ❖ A human questioner cannot tell if
 - ❖ There is a computer or a human answering his question, via teletype (remote communication).
 - ❖ The computer must behave intelligently.

Intelligent Behavior

- ❖ To achieve human-level performance in all cognitive tasks

What is Artificial Intelligence?



The Total Turing Test :

- Includes two more issues:

- 1. Computer vision* - To perceive objects (seeing)
- 2. Robotics* - To move objects (acting)

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Systems that think like humans - Cognitive Modeling:

- ❖ **Goal:** Make machines think the way humans actually think.
- ❖ An approach to AI that aims to build systems which not only produce human-like outputs, but also arrive at them through the same internal processes a human would use.
- ❖ The goal is to simulate the actual mechanics of human thought - perception, memory, attention, reasoning, problem-solving, inside a machine, not just imitate the result.
- ❖ It treats the human mind as an information-processing system and tries to reproduce it computationally.
- ❖ The cognitive modeling approach defines AI as systems that replicate the internal processes of human thought, not just the outputs.

What is Artificial Intelligence?

- ❖ It treats the human mind as the gold standard for intelligence and AI as a way to model and simulate it.
- ❖ How we know how humans think (sources of evidence):
 - ❖ **Introspection** — examining our own thought processes from the inside ("when I solve this puzzle, what steps do I follow?").
 - ❖ **Psychological experiments** — observing how people behave, react, and reason in controlled settings, then inferring underlying mental processes.
 - ❖ **Brain imaging and neuroscience** — studying the physical substrate of thought to ground the model in biology.

What is Artificial Intelligence?

❖ Connection to Cognitive Science:

- ❖ Cognitive modeling sits at the intersection of **AI, psychology, linguistics, neuroscience, and philosophy of mind.**
- ❖ AI provides the computational language; psychology and neuroscience provide the data; together they form Cognitive Science.
- ❖ Haugeland's framing: building "machines with minds in the full and literal sense."
- ❖ Bellman's framing: "the automation of activities we associate with human thinking — decision-making, problem-solving, learning."

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Systems that think 'rationally' - "Laws of Thought":

- ❖ **Goal:** Make machines reason using correct, logically valid steps.
- ❖ Roots go back to Aristotle's syllogisms — formal patterns of valid reasoning.
- ❖ Intelligence is defined as the ability to reason correctly - that is, to draw valid conclusions from given facts using the laws of logic.
- ❖ If we can encode knowledge as logical statements and apply formal rules of inference, then a machine should be able to "think" and solve problems by mechanical deduction.

What is Artificial Intelligence?

- ❖ The aim is not to imitate how humans actually think (which is often biased or flawed) but how they ought to think to reach truth.
- ❖ Knowledge about the world is expressed in a formal language.
- ❖ An inference engine systematically applies logical rules to derive new facts.
- ❖ Given enough correct premises, the system can prove theorems, answer queries, or plan actions.
- ❖ The "laws of thought" approach treats AI as applied logic. It is powerful where the world can be cleanly formalized, but breaks down in messy, uncertain, real-world settings - which is why modern AI complements logic with probability, learning, and the broader rational agent approach.

What is Artificial Intelligence?



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Systems that act rationally – "Rational Agent":

- ❖ **Goal:** Build agents that do the right thing — the action expected to achieve the best outcome given what is known.
- ❖ **Rational behavior:** Doing the right thing
- ❖ **The right thing:** That which is expected to maximize goal achievement, given the available information.
- ❖ **Definition of a rational agent**
 - ❖ An entity that perceives its environment through sensors and acts upon it through actuators.
 - ❖ It chooses the action expected to produce the best outcome, given everything it currently knows.
 - ❖ The agent is judged by what it does, not by how it thinks.

What is Artificial Intelligence?

❖ "Doing the right thing"

- ❖ "Right thing" = the action expected to maximize the performance measure.
- ❖ Performance measure is the external criterion of success (e.g., distance travelled safely, games won, profit earned).
- ❖ Rationality is evaluated on expected outcomes given available information, not perfect hindsight.

❖ What rationality is NOT

- ❖ Not the same as omniscience — a rational agent can still fail due to incomplete information; it just must have made the best choice with what it had.
- ❖ Not the same as pure logical reasoning — logic cannot express uncertainty, vagueness, or partial knowledge.

What is Artificial Intelligence?

- ❖ Not the same as thinking like a human — the agent need not replicate human reasoning or feelings.
- ❖ Not the same as always being correct — bad luck or bad information can lead to bad outcomes from rational choices.
- ❖ **Why "act rationally" is the dominant approach**
 - ❖ More general than logic alone — it incorporates probability, decision theory, and learning.
 - ❖ Combines LOGIC + domain knowledge + uncertainty handling.
 - ❖ Mathematically rigorous — built on probability theory and utility theory.
 - ❖ Doesn't require us to understand human psychology.
 - ❖ Compatible with scientific methodology: hypotheses, experiments, measurable performance

What is Artificial Intelligence?



- ❖ **Artificial Intelligence** is composed of two words **Artificial** and **Intelligence**, where **Artificial** defines "**man-made,**" and **Intelligence** defines "**thinking power**", hence AI means "a man-made thinking power."

What is Artificial Intelligence?



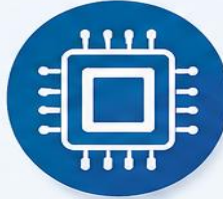
ARTIFICIAL

Produced by human art or effort, rather than originating naturally.



EXAMPLE

A bridge is artificial because it is designed and built by humans.



INTELLIGENCE

The ability to acquire knowledge and use it.



"The ability to acquire knowledge and use it"
— Pigford and Baur



EXAMPLE

Humans learn from experience and apply that knowledge to solve problems.



AI IS DEFINED AS:

- ✓ AI is the **study of ideas** that enable computers to be intelligent.
- ✓ AI is the part of computer science concerned with **design of computer systems that exhibit human intelligence.**
- ✓ (From the Concise Oxford Dictionary)



EXAMPLE

AI systems, like recommendation engines or virtual assistants, are designed to think, learn, and make decisions.



IN SIMPLE TERMS

AI is about creating machines that can think, learn, reason, and act in ways that normally require human intelligence.

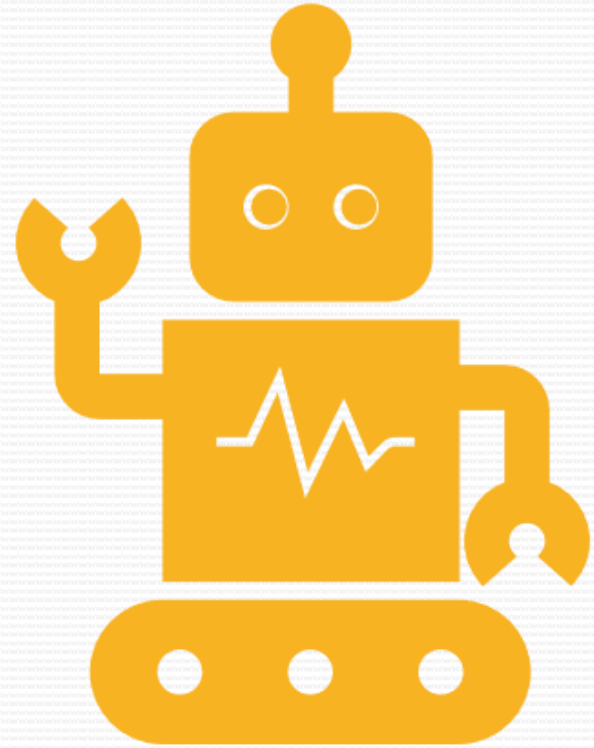
What is Artificial Intelligence?

❖ Therefore, we define AI as:

- ❖ A branch of computer science by which we can create intelligent machines which can behave like a human, think like humans, and able to make decisions.
- ❖ A branch of science which deals with helping machines find solutions to complex problems in a more human-like fashion.
- ❖ A branch of computer science that is concerned with the automation of intelligent behaviour.

What is Artificial Intelligence?

- ❖ From the above definitions, we can see that AI has two major roles:
 - ❖ Study the intelligent part concerned with humans.
 - ❖ Represent those actions using computers.



Features of AI

- ❖ AI is a technology that aims at replicating human intelligence. Some key features of AI are:
 1. **Ability to learn:** AI systems can improve their performance eventually by learning from data and past experiences.
 2. **Logical decision making:** AI systems are fed with large amounts of data to understand and recognize patterns for analysis and decision making.
 3. **Adaptability:** AI systems can adjust and adapt to changes in data.
 4. **Efficient automation:** AI would efficiently execute repetitive tasks and processes.
 5. **Versatility:** AI can be widely applied for various tasks across all fields like businesses, automotive, health, and many others.

History of AI

- ❖ Artificial Intelligence is not a new word and not a new technology for researchers.
- ❖ This technology is much older than you would imagine.
- ❖ Even there are the myths of Mechanical men in Ancient Greek and Egyptian Myths.
- ❖ The following are some milestones in the history of AI, tracing its journey from its earliest origins to the present day:

THE HISTORY OF ARTIFICIAL INTELLIGENCE

1936

Alan Turing introduces the concept of a universal machine.

The Turing Test

Where a human evaluator engages in natural language conversation with a machine and a human.

1943

McCulloch & Pitts publish a paper titled 'A Logical Calculus of Ideas Immanent in Nervous Activity', proposing the groundwork for neural networks.

1969

Shakey the Robot, one of the first general-purpose mobile robots, is developed at Stanford Research Institute (SRI).

1980

Expert systems gain popularity, with companies using them for financial forecasting and medical diagnoses.

DeepMind

Was acquired by Google in 2014 for \$500 million

2015

AlphaGo, developed by DeepMind, defeats world champion Lee Sedol in the game of Go.

2021

DeepMind's AlphaFold2 solves the protein-folding problem, paving the way for new drug discoveries and medical breakthroughs.

The Turing Test

Where a human evaluator engages in natural language conversation with a machine and a human.

1974

The first AI winter begins, marked by a decline in funding and interest in AI research due to unrealistic expectations and limited progress.

Neural Networks

Machine learning models that mimic the brain, learning to recognize patterns and make predictions via artificial neuron connections.

1950

Turing publishes 'Computing Machinery and Intelligence', proposing the Turing Test as a way to measure a machine's ability.

1951

Marvin Minsky and Dean Edmonds build SNAR, the first neural network computer.

1967

Newell and Simon develop the General Problem Solver (GPS), one of the first AI programs to demonstrate human-like problem-solving.

1986

Hinton, Rumelhart, and Williams publish 'Learning Representations by Back-Propagating Errors', allowing much deeper neural networks to be trained.

2006

Deep learning gains momentum with breakthroughs in training large-scale neural networks.

2009

Google Brain project begins, focusing on deep learning research at scale.

2011

IBM's Watson defeats two former Jeopardy! champions.

2012

AI startup DeepMind develops a deep neural network that can recognize cats in YouTube videos.

2014

Facebook creates DeepFace, a facial recognition system that can recognise faces with near-human accuracy.

2016

DeepMind's AlphaGo defeats world champion Lee Sedol 4-1 in a historic Go rematch.

2017

Google's AlphaZero defeats the world's best chess, shogi, and Go engines in a series of matches.

2018

ELMo by AllenNLP improves natural language understanding with deep contextual word representations.

2019

OpenAI releases GPT-2, a powerful unsupervised language model.

2020

OpenAI releases GPT-3, marking a significant breakthrough in natural language processing.

2022

Google fires engineer Blake Lemoine over his claims that Google's Language Model for Dialogue Applications (LaMDA) was sentient.

2023

Artists file a class-action lawsuit against Stability AI, DeviantArt, and Midjourney for their use of Stable Diffusion to remix the copyrighted works of millions of artists.

2024

OpenAI launches Sora, a text-to-video model capable of generating highly realistic videos from text prompts.

2025

AI agents become mainstream, assisting with complex tasks across work, education, and daily life.

The Dartmouth Conference (organised by McCarthy, Minsky, Rochester, and Shannon) is held, marking the birth of AI as a field of study.

1957

Rosenblatt develops the Perceptron: the first artificial neural network capable of learning.

1965

Weizenbaum develops ELIZA: a natural language processing program that simulates conversation.

Natural Language Processing

Teaches computers to understand and use human language using techniques like machine learning.

History of AI

Maturation of Artificial Intelligence (1943-1952)

- ❖ **1943:** The first work which is now recognized as AI was done by Warren McCulloch and Walter Pitts in 1943. They proposed a model of artificial neurons.
- ❖ **1949:** Donald Hebb demonstrated an updating rule for modifying the connection strength between neurons. His rule is now called Hebbian learning.
- ❖ **1950:** Alan Turing publishes "Computing Machinery and Intelligence" in which he proposed a test. The test can check the machine's ability to exhibit intelligent behavior equivalent to human intelligence, called a Turing test.

History of AI

The birth of Artificial Intelligence (1952-1956).

- ❖ **1955:** Allen Newell and Herbert A. Simon created the "first artificial intelligence program" which was named as "Logic Theorist". This program had proved 38 of 52 Mathematics theorems, and find new and more elegant proofs for some theorems.
- ❖ **1956:** The word "Artificial Intelligence" first adopted by American Computer scientist John McCarthy at the Dartmouth Conference. For the first time, AI coined as an academic field. At that time high-level computer languages such as FORTRAN, LISP, or COBOL were invented. And the enthusiasm for AI was very high at that time.

History of AI

The golden years-Early enthusiasm (1956-1974).

- ❖ **1966:** The researchers emphasized developing algorithms which can solve mathematical problems. Joseph Weizenbaum created the first chatbot in 1966, which was named as ELIZA.
- ❖ **1972:** The first intelligent humanoid robot was built in Japan which was named as WABOT-1.

History of AI

The first AI winter (1974-1980).

- ❖ The duration between years 1974 to 1980 was the first AI winter duration. AI winter refers to the time period where computer scientist dealt with a severe shortage of funding from government for AI researches.
- ❖ During AI winters, an interest of publicity on artificial intelligence was decreased.

History of AI

A boom of AI (1980-1987).

- ❖ **1980:** After AI winter duration, AI came back with "Expert System". Expert systems were programmed that emulate the decision-making ability of a human expert. Also, the first national conference of the American Association of Artificial Intelligence was held at Stanford University.

History of AI

The second AI winter (1987-1993)

- ❖ The duration between the years 1987 to 1993 was the second AI Winter duration.
- ❖ Again, investors and governments stopped funding AI research due to high costs and disappointing results. Expert systems like XCON, once highly profitable, became expensive and difficult to maintain, scale, and update, leading to the collapse of the LISP machine market.

History of AI

Shift to "Intelligent Agents" (1993–2011)

- ❖ **Web Browsing and Search:** AI was utilized to improve search engines and create agents that could navigate the rapidly expanding internet for information.
- ❖ **Data Mining and Commercial Use:** Businesses began using AI for data analysis, logistics, and, by the late 90s, early recommendation systems, often without calling it "AI".
- ❖ **Rise of Machine Learning:** Researchers began focusing on statistical methods, allowing systems to learn from data rather than relying solely on pre-defined, hand-coded rules.

History of AI

- ❖ **Bayesian Networks:** Probabilistic reasoning became a dominant approach, allowing AI to handle uncertainty and incomplete information, a major improvement over the rigid expert systems of the 1980s.
- ❖ **Hardware Improvements:** Increased computer processing power and memory capacity allowed for more complex, practical applications.
- ❖ **1997: IBM's Deep Blue:** The most significant event of the decade was IBM's chess-playing supercomputer, Deep Blue, defeating reigning world champion Garry Kasparov in a six-game match. This demonstrated that AI could achieve complex, strategic decision-making.

History of AI

- ❖ **1997: Speech Recognition:** Dragon Systems released "NaturallySpeaking," the first commercial speech recognition software for Windows, making voice input a reality.
- ❖ **1997: Long Short-Term Memory (LSTM):** Jürgen Schmidhuber and Sepp Hochreiter proposed the LSTM architecture, a type of recurrent neural network that became fundamental for modern speech and handwriting recognition.
- ❖ **1998: Kismet:** MIT researcher Cynthia Breazeal developed Kismet, an expressive, humanoid robot designed to recognize and simulate human emotions, advancing human-robot interaction.
- ❖ **1999: Sony's AIBO:** Sony released AIBO, a robotic dog capable of learning, interacting with its environment, and responding to over 100 voice commands, bringing AI into the consumer market.

History of AI

Deep Learning, Big Data And Artificial General Intelligence (2011-Present)

- ❖ **2011:** IBM's Watson won Jeopardy!, a quiz show, where it had to solve the complex questions as well as riddles.
Watson had proved that it could understand natural language and can solve tricky questions quickly.
- ❖ **2012:** Google has launched an Android app feature "Google now", which was able to provide information to the user as a prediction.
- ❖ **2014:** 2014: The chatbot "Eugene Goostman" reportedly passed the Turing Test by convincing 33% of judges it was a 13-year-old boy (though the claim remains contested).

History of AI

- ❖ **2015:** Baidu's Minwa supercomputer uses a special deep neural network called a convolutional neural network to identify and categorize images with a higher rate of accuracy than the average human.
- ❖ **2016:** DeepMind's AlphaGo program, powered by a deep neural network, beats Lee Sedol, the world champion Go player, in a five-game match. The victory is significant given the huge number of possible moves as the game progresses (over 14.5 trillion after just four moves). Later, Google purchased DeepMind for a reported USD 400 million.

History of AI

- ❖ **2018:** The "Project Debater" from IBM debated on complex topics with two master debaters and also performed extremely well.
- ❖ **2022:** A rise in large language models or LLMs, such as OpenAI's ChatGPT, creates an enormous change in performance of AI and its potential to drive enterprise value. With these new generative AI practices, deep-learning models can be pretrained on large amounts of data.

History of AI

- ❖ **2024:** The latest AI trends point to a continuing AI renaissance.
- ❖ Multimodal models that can take multiple types of data as input are providing richer, more robust experiences. These models bring together computer vision image recognition and NLP speech recognition capabilities. Smaller models are also making strides in an age of diminishing returns with massive models with large parameter counts.
- ❖ Multimodal models (e.g., GPT-4o, Gemini 1.5, Claude 3.5) became mainstream, handling text, images, audio, and video in a single model. The EU AI Act was passed, the first comprehensive law regulating AI.

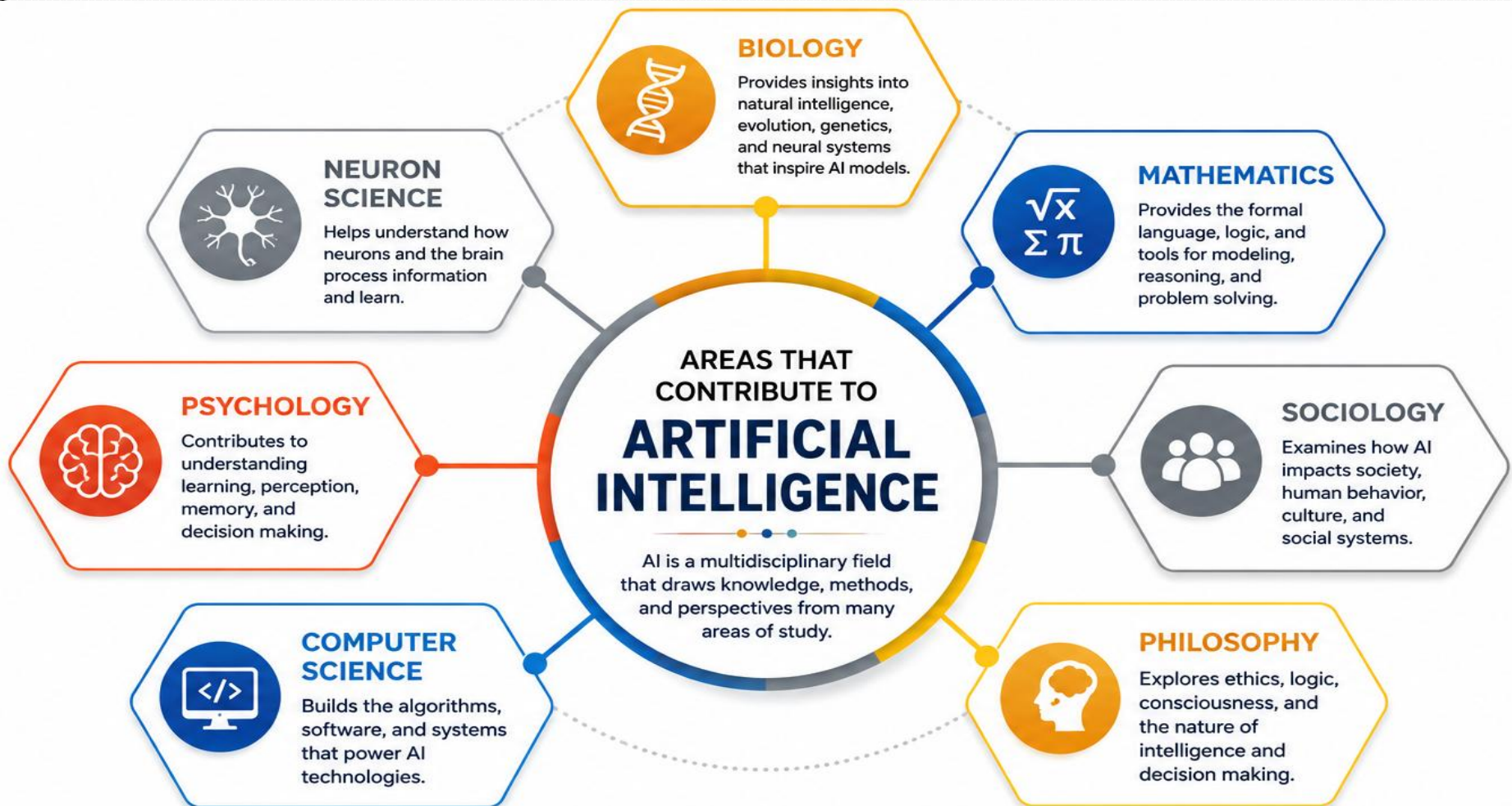
History of AI

- ❖ **2025:** The rise of reasoning models (OpenAI o1/o3, Claude with extended thinking, DeepSeek R1) which “think” step-by-step before answering, achieving expert-level performance in mathematics, science, and coding.
- ❖ **2025–2026:** The shift to Agentic AI: autonomous AI agents that browse, code, and use computers on a user’s behalf (e.g., Claude Code, Operator, Computer Use). Open-source models close the gap with frontier labs.
- ❖ Google has demonstrated an AI program "Duplex" which was a virtual assistant, and which had taken hairdresser appointment on call, and lady on other side didn’t notice that she was talking with the machine.

History of AI

- ❖ Now AI has developed to a remarkable level. The concept of Deep learning, big data, and data science are now trending like a boom.
- ❖ Nowadays companies like Google, Meta (Facebook), IBM, Microsoft, OpenAI, Anthropic, and Amazon are working with AI and creating remarkable products.
- ❖ The future of Artificial Intelligence is inspiring and will come with high intelligence.

Foundations of AI



STRONGER TOGETHER

AI advances when diverse fields collaborate, combining knowledge to create smarter, more human-centered technologies.

Foundations of AI

❖ Philosophy

- ❖ AI's origins are deeply rooted in philosophical questions about the nature of knowledge, learning, and reasoning.
- ❖ Early philosophical inquiries laid the groundwork for understanding the mind and how it processes information.
- ❖ Initiate the idea of mind as a machine and its internal operations.

By: Ambrose Njeru [BSc, MSc.]



Foundations of AI

- ❖ **Mathematics** - Formalizes the three main area of AI: Computation, Logic, and Probability.
- ❖ Computation leads to analysis of the problems that can be computed (Complexity theory).
- ❖ Probability contributes the “degree of belief” to handle uncertainty in AI.
- ❖ Decision theory combines probability theory and utility theory (bias).



Foundations of AI

❖ Psychology

- ❖ How do humans think and act?
- ❖ The study of human reasoning and acting.
- ❖ Provides reasoning models for AI
- ❖ Strengthen the ideas.
 - ❖ Humans and other animals can be considered as information processing machines

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Foundations of AI

❖ Computer Science

- ❖ How to build an efficient computer?
- ❖ Computer science provides the tools and techniques for implementing AI concepts.
- ❖ The power of computer makes computation of large and difficult problems more easily
- ❖ AI has also contributed its own work to computer science, including: time- sharing, the linked list data type, OOP, etc.

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Foundations of AI

- ❖ **Neuroscience.**
 - ❖ Understanding how the human brain works has influenced AI development, particularly in areas like neural networks.
 - ❖ Artificial neural networks, which mimic the structure and function of biological neurons, are the basis for deep learning and modern AI.

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Foundations of AI

- ❖ **Control Theory and Cybernetics**
 - ❖ How can artifacts operate under their own control?
 - ❖ The artifacts adjust their actions
 - ❖ To do better for the environment over time.
 - ❖ Based on an objective function and feedback from the environment.
 - ❖ Not limited only to linear systems but also other problems such as language, vision, & planning, etc.

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Foundations of AI

❖ Linguistics

- ❖ Natural language processing (NLP) relies on linguistic principles to enable AI systems to understand and generate human language.
- ❖ NLP is a key component of AI, allowing machines to communicate and interact with humans more effectively.
- ❖ Syntactic and semantic analysis.
- ❖ Knowledge representation.



Foundations of AI

❖ Algorithms

- ❖ AI system can assess data and draw conclusions from it because of algorithms, which are detailed instructions or sets of rules.
- ❖ Algorithms have become more and more advanced, enabling AI to perform incredibly difficult jobs like language translation and image identification.
- ❖ Algorithms provide the basis of AI, allowing it to learn from data and make predictions.

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Major Schools of AI

- ❖ **Major schools of AI** - How they reshape the way we interact with machines and understand artificial intelligence.



Major Schools of AI

- ❖ *“Artificial intelligence is the science of making machines do things that would require intelligence if done by men.” - Marvin Minsky*
- ❖ Within the expansive realm of AI, various schools of thought have emerged, each offering unique perspectives on how machines can simulate human-like cognitive abilities.
- ❖ The major schools of AI are:
 - ❖ Symbolism.
 - ❖ Connectionism.
 - ❖ Actionism.

Major Schools of AI

1. Major schools of AI - Symbolism:

- ❖ **Symbolism**, also known as **logicism**, **psychologism**, or **computerism**, is a branch of artificial intelligence that deals with symbolic representation derived from mathematical logic.
- ❖ It is a school of thought that believes the world can be explained by symbols and dealt with through precise, logical processes.

Major Schools of AI

2. Major schools of AI - Connectionism:

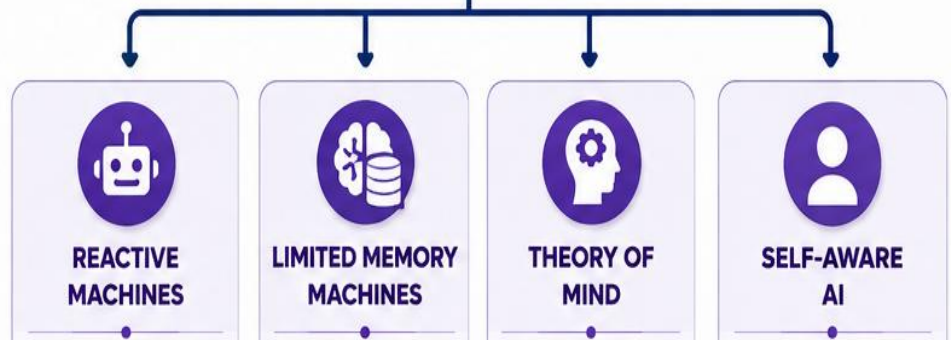
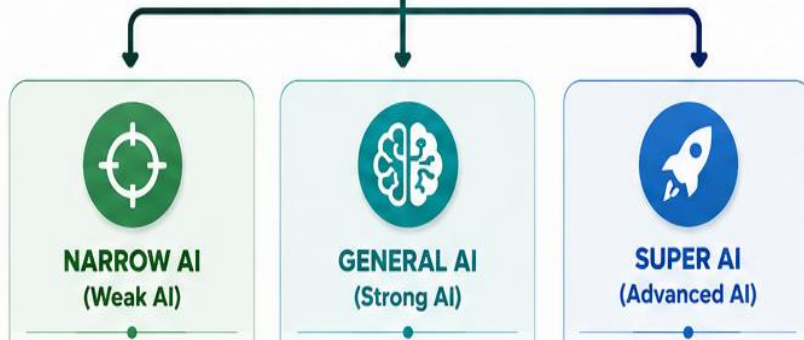
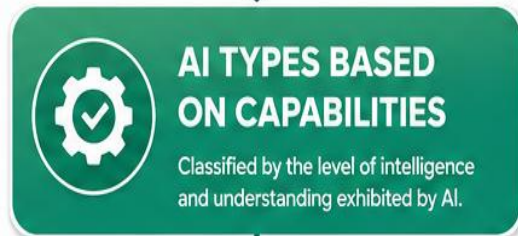
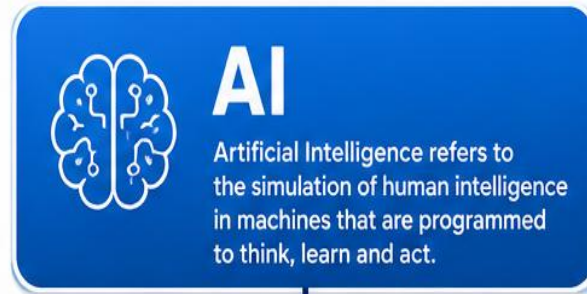
- ❖ **Connectionism**, on the other hand, known as **bionicsism** or **physiologism**, believes that this process should be implemented through artificial neural networks.
- ❖ It is an approach to the study of human mental processes and cognition that utilizes mathematical models known as connectionist networks or artificial neural networks.
- ❖ Connectionist networks are composed of simple and often uniform units that are interconnected.
- ❖ The connections between these units can vary from model to model.

Major Schools of AI

3. Major schools of AI - Actionism:

- ❖ **Actionism** is also known as **evolutionism** or **cyberneticsism**, posits that AI originates from cybernetics.
- ❖ It proposes that intelligence is contingent on perception and actions, asserting that intelligence doesn't necessitate knowledge, representation, or reasoning.
- ❖ According to this perspective, AI has the capacity to evolve akin to human intelligence, and intelligent behavior can only manifest in the real world through interaction with the surrounding environment.

Types of AI



AI Types by Capabilities

- ❖ AI, can be categorized into three main types based on their capabilities on how they learn and how far they can apply their knowledge:
 - ❖ Narrow AI (also known as Weak AI).
 - ❖ General AI (also known as Strong AI)
 - ❖ Super AI.

1. **Narrow AI (Weak AI)**

- ❖ This type of AI is designed to perform specific narrow tasks (e.g., facial recognition, internet searches, or driving a car).
- ❖ Most current AI systems, fall under this category.

AI Types by Capabilities

- ❖ They are built to serve and excel in one cognitive capability and cannot independently learn skills beyond their design.
- ❖ They operate under a limited pre-defined range or set of contexts.
- ❖ Utilizes machine learning, NLP and neural network algorithms to complete specified tasks.
- ❖ **Characteristics:** Can't generalize to other tasks or learn beyond its specific programming.
- ❖ **Examples:** Virtual assistants (Siri, Alexa), image recognition software, recommendation engines.

AI Types by Capabilities

2. General AI (Strong AI)

- ❖ A type of AI endowed with broad human-like cognitive capabilities, enabling it to tackle new and unfamiliar tasks autonomously.
- ❖ Hypothetical AI with human-level intelligence, able to perform any intellectual task a human can.
- ❖ Such a robust AI framework possesses the capacity to discern, assimilate, and utilize its intelligence to resolve any challenge without needing human guidance.
- ❖ **Current status:** Not yet developed.

AI Types by Capabilities

- ❖ **Characteristics:** Possesses consciousness, self-awareness, and the ability to make independent decisions, which are not currently achievable.

3. Super AI

- ❖ This represents a future form of AI where machines could surpass human intelligence across all fields, including creativity, general wisdom, and problem-solving.
- ❖ Superintelligence is speculative and not yet realized.
- ❖ **Characteristics:** Can surpass human problem-solving, creativity, and overall abilities.

AI Types by Functionalities

- ❖ AI by functionality is based on its interaction with environment.
- ❖ Current AI systems can perform complicated computations at high speed.
- ❖ They can manage large amounts of data and generate reliable predictions.
- ❖ Artificial intelligence is divided into four levels:

1. Reactive Machines

- ❖ These are the machines that have no memories or previous experiences.
- ❖ It is 'reactive' in nature.

AI Types by Functionalities

- ❖ AI's successful implementation shows that in the late 1990s, IBM's Deep Blue chess computer defeated international grandmaster Garry Kasparov six times in a row.
- ❖ Deep Blue could recognize chess pieces and understand how they should move to win.
- ❖ Its great intelligence allows it to guess all of the opponent's probable movements faster than a human.
- ❖ As a consequence, it could compute the best movements for each scenario.

AI Types by Functionalities

2. **Limited Memory:** This sort of contraption can see back in time. A famous example is self-driving automobiles which can detect the speed and direction of other vehicles.
3. **Theory of Mind:** This classification contains machine learning, which attempts to reproduce the entire physical world - humans, animals, and the things that can think and feel.
4. **Self-awareness:** This class has machines with self-aware systems. This level extends the Theory of Mind stage by giving robots self-awareness for a "reason." This will give machines new intelligence.

Branches of AI

- ❖ Artificial Intelligence (AI) is a broad field that encompasses various subfields and branches, each focusing on different aspects of creating intelligent machines.



Branches of AI

1. Machine Learning:

- ❖ Machine learning (ML) is the science of empowering machines to make decisions without human intervention.
- ❖ This sub-discipline forms the backbone of AI, enabling computers to learn and interpret patterns in images, sounds, and structured data using multidimensional arrays.
- ❖ ML is further subdivided into four types of learning:



MACHINE LEARNING

Machine learning enables systems to learn from data, improve with experience, and make decisions or predictions without being explicitly programmed.

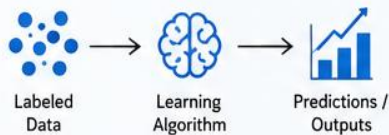


SUPERVISED

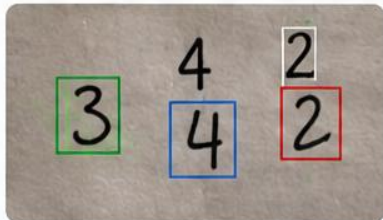
Learning with labeled data

Generate predictions or make decisions by learning from input-output pairs in **labeled datasets**.

HOW IT WORKS



EXAMPLE APPLICATION



Handwriting Recognition

- ✓ Recognize handwritten digits or text
- ✓ Used in ZIP code recognition, bank check processing, form digitization
- ✓ High accuracy with labeled examples

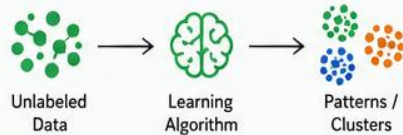


UNSUPERVISED

Finding patterns in unlabeled data

Find hidden structures, patterns, or relationships in data without using any **labeled responses**.

HOW IT WORKS



EXAMPLE APPLICATION



Geospatial Market Segmentation

- ✓ Group locations based on similar characteristics
- ✓ Help businesses identify target markets and optimize strategies
- ✓ No prior labels required

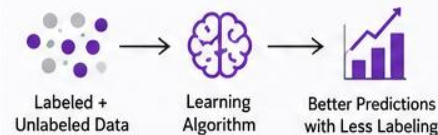


SEMI-SUPERVISED

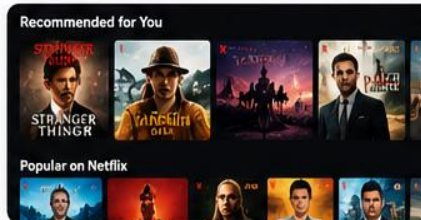
Learning with a little labeled data

Improve learning accuracy by using a **small amount of labeled data** together with a large amount of unlabeled data.

HOW IT WORKS



EXAMPLE APPLICATION



Interactive Recommendations

- ✓ Recommend movies, products, or content
- ✓ Leverage limited user feedback with large behavioral data
- ✓ Enhances personalization at scale



REINFORCEMENT

Learning by trial and reward

Train agents to make decisions by interacting with an environment and receiving **rewards or penalties** over time.

HOW IT WORKS



EXAMPLE APPLICATION



Self-Driving Vehicles

- ✓ Learn to drive safely in complex environments
- ✓ Make real-time decisions (e.g., steer, brake, accelerate)
- ✓ Improve performance with experience

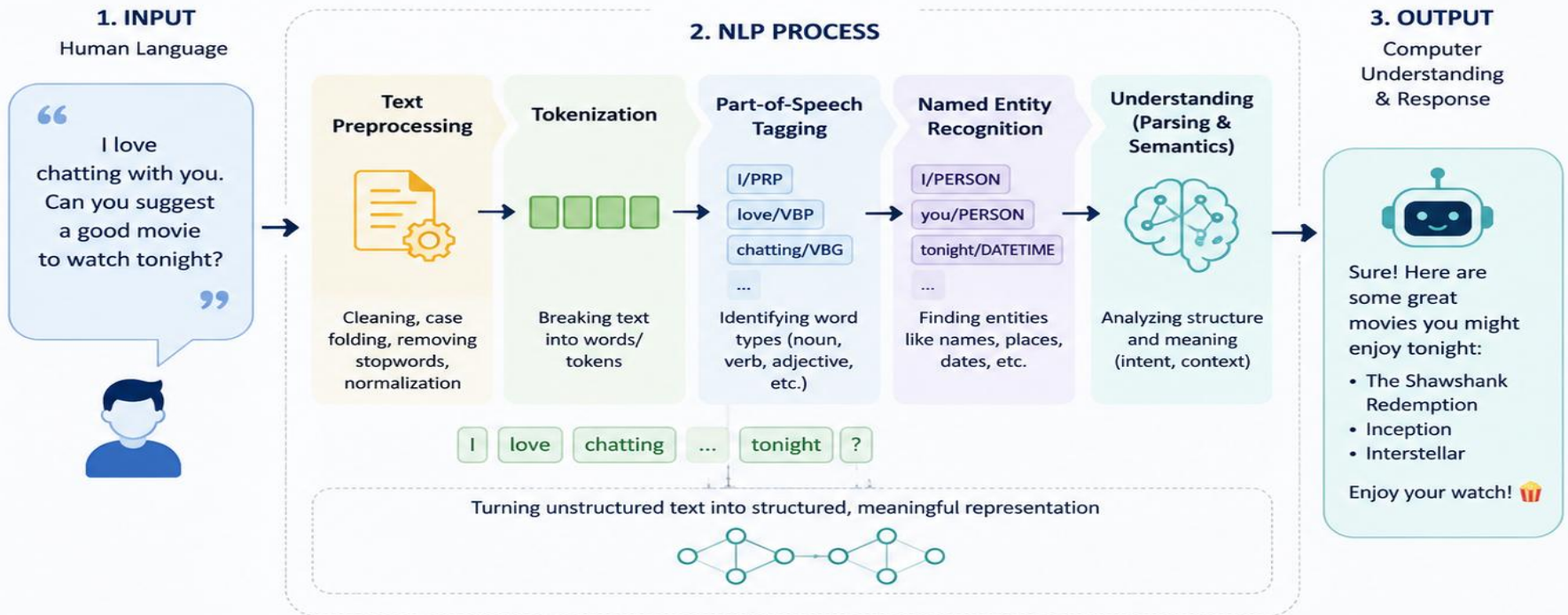


Key Takeaway: Each machine learning approach solves different problems. Choosing the right type depends on the data available, the goal, and how the model will interact with the world.

Branches of AI

2. Natural Language Processing:

Enabling computers to understand, interpret and generate human language



4. APPLICATIONS OF NLP


Chatbots & Virtual Assistants


Machine Translation


Information Extraction


Sentiment Analysis


Text Summarization


Speech Recognition


Text Generation

Branches of AI

3. Computer Vision:

- ❖ Computer vision (CV) is the automatic extraction, analysis, and interpretation of images or videos.
- ❖ CV converts photos and videos into numerical arrays, enabling ML algorithms to draw inferences, make predictions, and even generate new images based on user-defined inputs.
- ❖ Potential uses for CV have been studied for decades, but CV has only recently become possible at scale thanks to three innovations:

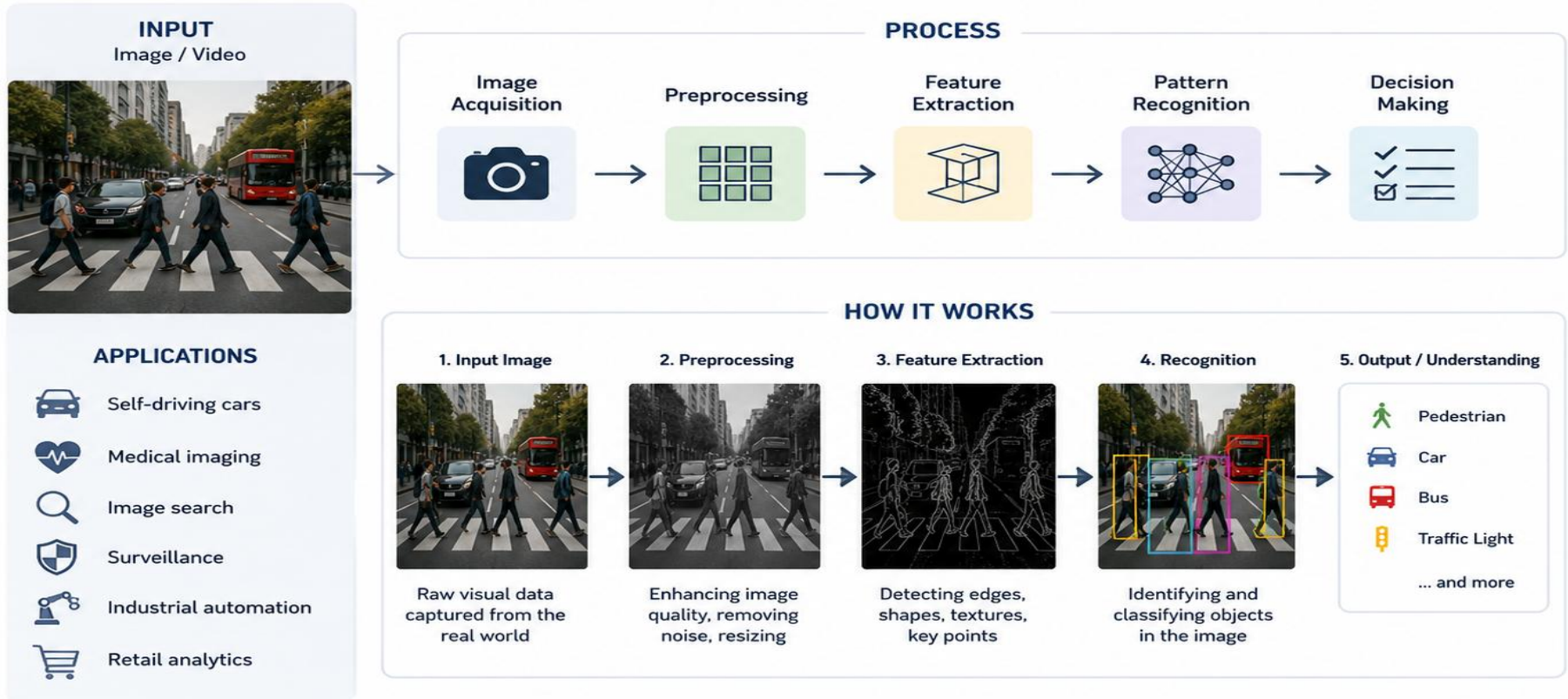
Branches of AI

- ❖ **More efficient algorithms:** Deep learning and convolutional neural networks, specifically significantly reduces the memory footprint and computational runtime of CV tasks.
- ❖ **Better computing resources:** GPU improvements, distributed architectures (e.g., Spark), and the availability of inexpensive cloud computing resources have made it cheaper than ever to run memory-hungry CV algorithms.
- ❖ **Availability of images to train on:** The proliferation of social media platforms, community forums, and digital / mobile cameras have drastically increased the number of publicly-available images that can be used to train CV algorithms.

Branches of AI

COMPUTER VISION

Enabling computers to see, understand and make decisions from visual data



TECHNOLOGIES BEHIND COMPUTER VISION


Deep Learning
(Convolutional Neural Networks)


Image Processing


Machine Learning


Big Data

Branches of AI

4. Robotics:

- ❖ Robotics is the science of designing, constructing, operating, and applying robots to solve human problems.
- ❖ Robots come in thousands of shapes and sizes, making it difficult to nail down the precise meaning of the term.
- ❖ Most robots rely on expert systems to accomplish their objectives.

Branches of AI

5. Expert Systems:

- ❖ An expert system (ES) is an artificial agent which leverages pre-programmed knowledge to offer advice or make decisions.
- ❖ An expert system can take advantage of human insights discovered through trial and error.
- ❖ Expert systems are more predictable and are less likely to make extreme errors when faced with previously-unseen inputs.
- ❖ Expert systems have historically been faster and easier to implement, though ML has become much more accessible in recent years.

Branches of AI

6. Deep Learning:

- ❖ Deep learning is a subset of machine learning that uses multilayered neural networks, called deep neural networks, that more closely simulate the complex decision-making power of the human brain.
- ❖ Deep neural networks include an input layer, at least three but usually hundreds of hidden layers, and an output layer, unlike neural networks used in classic machine learning models, which usually have only one or two hidden layers.

Branches of AI

7. Fuzzy Logic:

- ❖ Fuzzy Logic is a form of many-valued logic that deals with reasoning that is approximate rather than fixed and exact.
- ❖ It allows for variables to have truth values between 0 and 1, representing degrees of truth rather than just true or false.
- ❖ It's used to handle uncertainty and imprecision, making it useful in areas like control systems, image processing, and artificial intelligence.

Branches of AI

8. Evolutionary Computation (EC):

- ❖ A computational framework inspired by the principles of biological evolution.
- ❖ It uses algorithms like genetic algorithms to solve optimization problems by iteratively refining a population of solutions through selection, crossover, and mutation.

Branches of AI

9. Swarm Intelligence:

- ❖ A decentralized, self-organized system where a group of simple agents interact to achieve complex tasks, mimicking the collective behavior of natural swarms like ant colonies or bird flocks.
- ❖ The approach draws inspiration from natural systems where individuals follow simple rules, yet the group as a whole exhibits sophisticated behavior.

Branches of AI

10. Cognitive Computing:

- ❖ A field of AI that aims to simulate human thought processes, using computerized models to interpret data, understand natural language, and learn from interactions.
- ❖ It differs from traditional AI in that it seeks to emulate human cognitive abilities like reasoning and decision-making, rather than simply executing specific tasks based on pre-defined algorithms.

Branches of AI

11. Generative AI:

- ❖ Generative AI, sometimes called "gen AI", refers to deep learning models that can create complex original content such as long-form text, high-quality images, realistic video or audio and more in response to a user's prompt or request.
- ❖ At a high level, generative models encode a simplified representation of their training data, and then draw from that representation to create new work that's similar, but not identical, to the original data.

Branches of AI

12. Large Language Models & Agentic AI:

- ❖ Large Language Models (LLMs) are foundation models trained on massive text corpora (e.g., GPT-4, Claude, Gemini, Llama). They power chatbots, code assistants, search, and writing tools, and have become the dominant interface for AI since 2022.
- ❖ Agentic AI extends LLMs with the ability to plan, use tools, and act autonomously to accomplish multi-step goals, browsing the web, writing and executing code, controlling computers, and coordinating with other agents on a user's behalf.

Goals of AI

- ❖ The primary goal of AI is to simulate human-like intelligence in machines, empowering them to carry out complex tasks and decision-making processes autonomously.
- ❖ The objectives of AI encompass a wide range of applications, including but not limited to:
 1. **Problem-Solving and Decision Making.**
- ❖ Develop systems that can analyze large datasets, identify patterns, and make data-driven decisions.
- ❖ Ability to solve problems and make decisions efficiently.

Goals of AI

2. Natural Language Processing (NLP).

- ❖ AI-driven NLP is a critical aspect of creating machines that can understand and communicate with humans in natural language.
- ❖ NLP enables virtual assistants like Siri to comprehend user queries and respond appropriately, making machine interactions more intuitive and user-friendly.

3. Robotics and Automation.

- ❖ Integrating AI with robotics has given rise to intelligent machines that can perform physical tasks with precision and accuracy.

Goals of AI

4. Machine Learning and Deep Learning.

- ❖ Subset of AI that focus on enabling machines to learn from data without explicit programming.
- ❖ The techniques have led to significant advancements in computer vision, speech recognition.

5. Enhancing Healthcare and Medicine.

- ❖ To improve diagnostics, treatment planning, and patient care.
- ❖ Leveraging AI algorithms to analyze medical images, predict disease outcomes, and develop personalized treatment plans for patients.

Goals of AI

6. Fostering Creativity and Innovation

- ❖ Potential to spur creativity and innovation.
- ❖ AI-powered tools can assist artists, writers, and designers generate creative new ideas and push the boundaries of human imagination.

7. Learning and Adaptation:

- ❖ AI focuses on developing systems that can learn from data and improve their performance over time. Includes utilizing machine learning techniques to enable systems to learn from past experiences and observations.
- ❖ AI systems can adapt to new information and become more accurate and efficient in their tasks.

AI Applications

- ❖ AI has various applications today and it is becoming essential for today's time because it can solve complex problems with an efficient way in multiple industries.



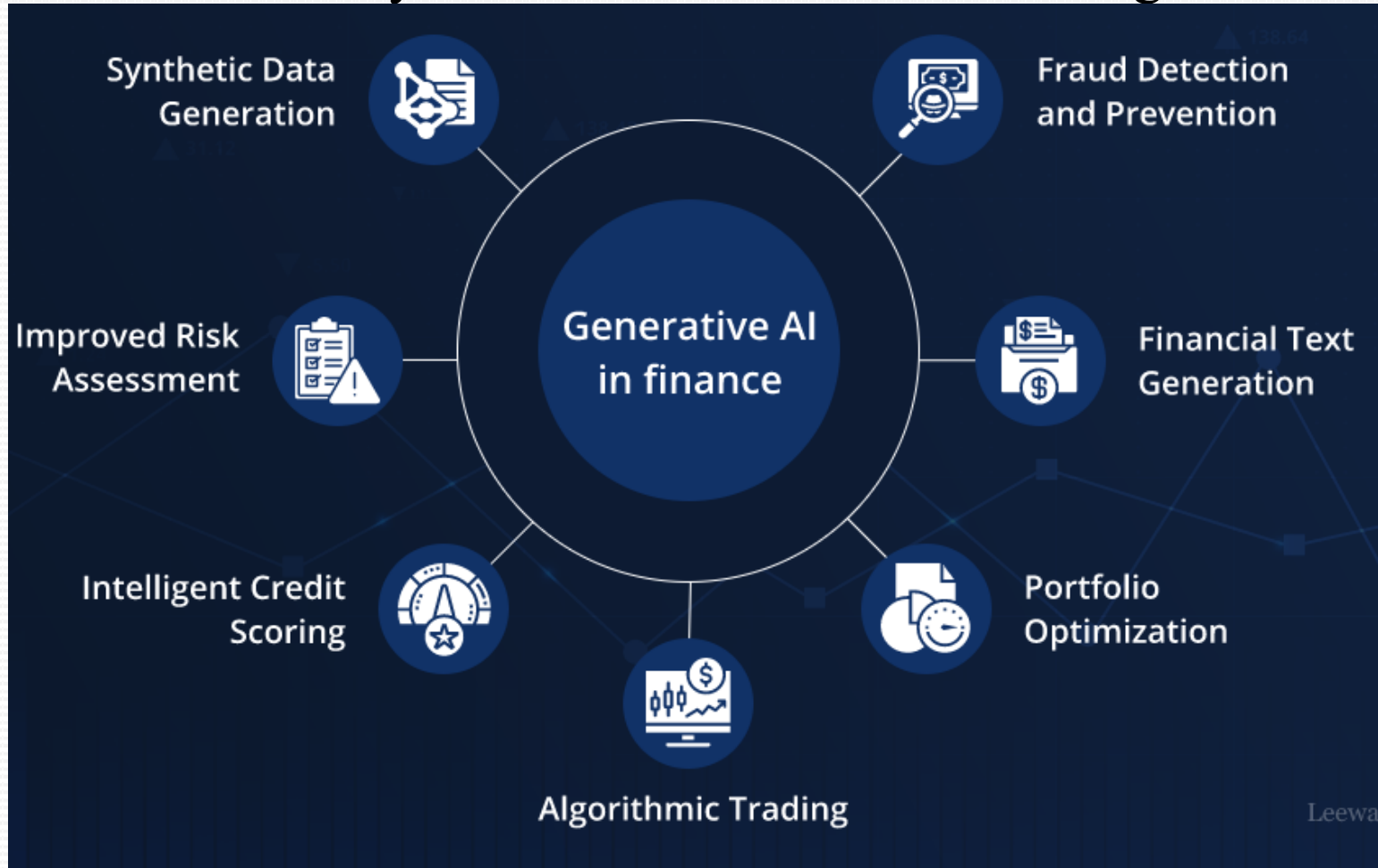
AI Applications

1. **Robotics:** Used in many industries.



AI Applications

2. **Finance:** Help to identify fraud, manage risk, make credit decision and analyze data to find details that might be missed.



AI Applications

3. **Healthcare:** Help doctors diagnose diseases, develop new treatments, and provide personalized care.



AI Applications

4. **Marketing:** Help marketers analyze psychology and recognize patterns to create personalized advertising campaigns.



AI Applications

5. **Education:** Create action plans for each student based on their data and capabilities.



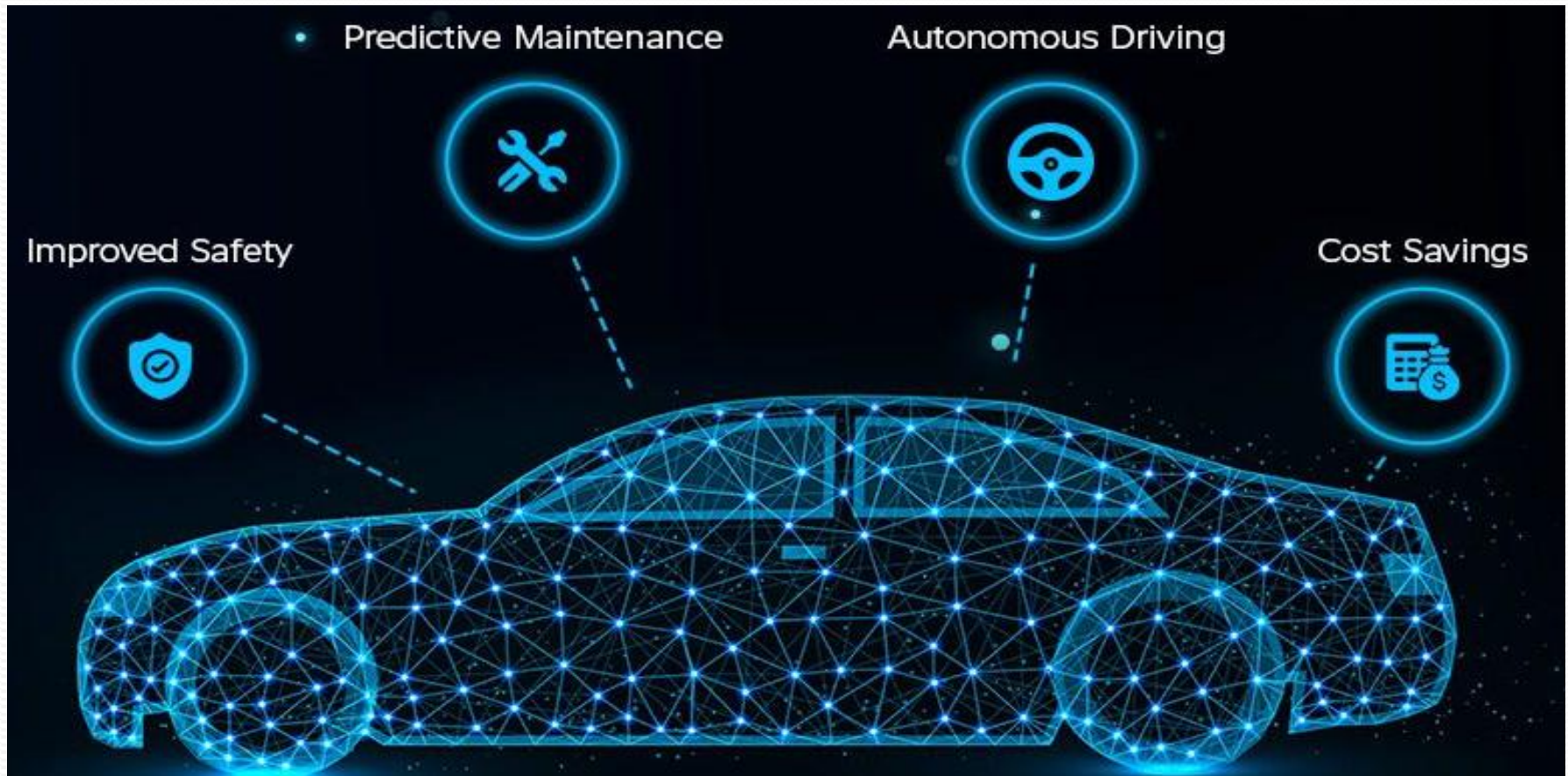
AI Applications

- 6. Natural language processing (NLP):** Helps computers understand, interpret, and generate human language.



AI Applications

7. **Automobiles:** Help build self-driving cars by running the car, along with its camera, radar, GPS, and more.



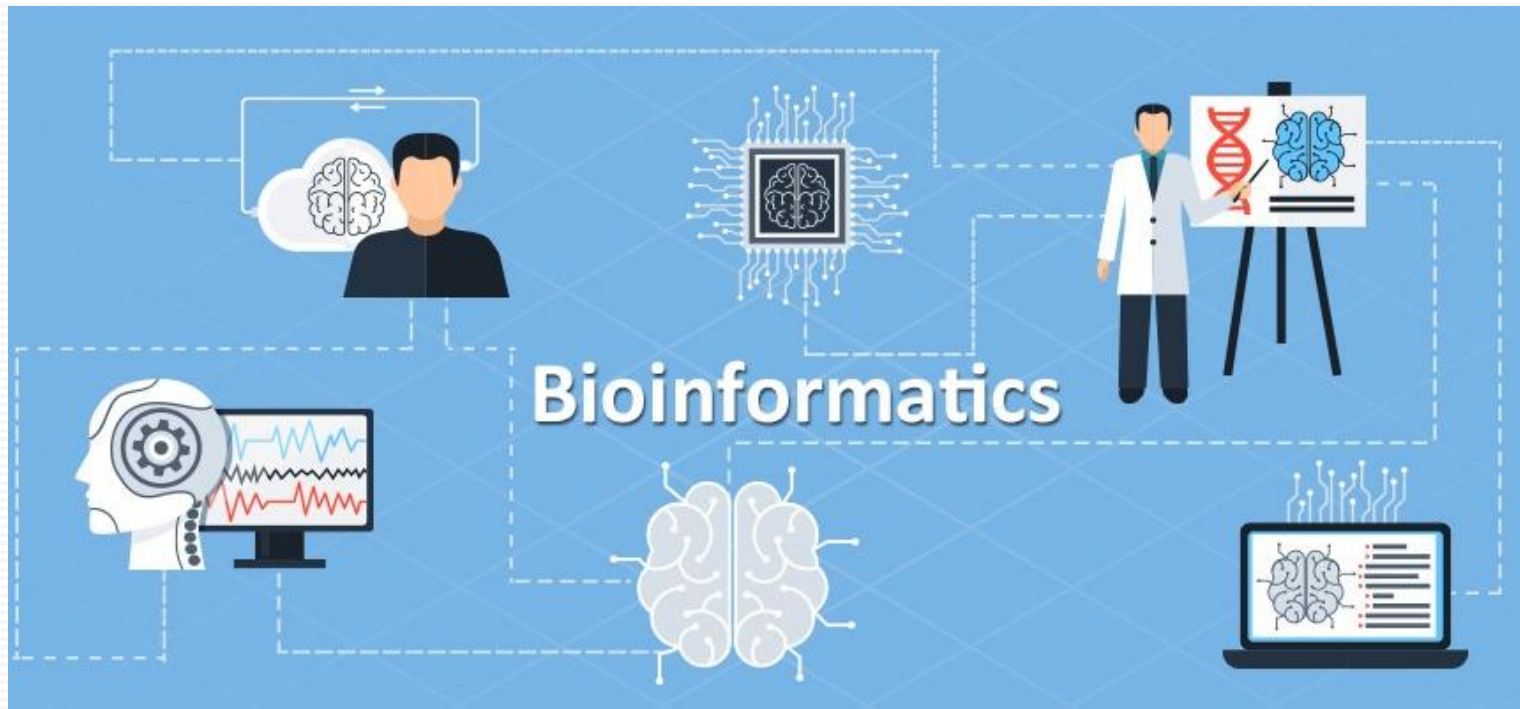
AI Applications

8. **Agriculture:** Help farmers choose when to plant seeds, what nutrients to apply, and how to space seeds.



AI Applications

9. **Bioinformatics:** Has many applications, including drug discovery, personalized medicine, and disease diagnosis. Can also be used to analyze large biological datasets to gain insights into population-level genetic variants.



AI Applications

10. Games: Character creation, predictive analysis, game analytics, voice recognition.



AI Applications

- 11. Data Security:** The security of data is crucial for every company and cyber-attacks are growing very rapidly in the digital world. AI can be used to make your data more safe and secure.



AI Advantages

1. More powerful and more useful computers.
2. Reduces human errors in repetitive and data-heavy tasks.
3. Increase productivity.
4. Takes on dangerous or risky tasks (e.g., disaster response, hazardous environments).
5. Helps in strategic decision making.
6. AI systems help reduce the cost of goods and services.
7. AI-based systems are able to handle repetitive tasks reliably.
8. Round-the-clock availability (24/7).
9. Handles low level tasks perfectly.
10. Creates new opportunities, jobs, and industries in the market.

AI Disadvantages

1. **Job displacement:** automation and generative AI are now affecting both manual and white-collar roles (e.g., writing, coding, customer service).
2. Poses an existential risk.
3. **Errors and hallucinations:** AI systems, especially LLMs, can produce confident but false outputs that require skilled human verification.
4. High cost to develop AI based system.
5. Sometimes not able to handle highly intelligent tasks, so human interaction needed.
6. A complex system to train datasets.
7. Military and weapons applications, including autonomous weapons, raise serious safety and ethical concerns.

AI Ethics

- ❖ Ethics in AI is a set of principles and considerations which enhance the development, deployment, and impact of AI technologies.
- ❖ The key ethical issues in AI include:
 1. **Privacy:** We feed the machines with the personal details of people to help it think and act like humans. But how do we know that the details are secure and private? Data privacy is one of the crucial concerns in the development and use of AI.
 2. **Transparency:** Refers to the practice of making AI systems and their operations understandable to users, and the way to do this is disclosure.

AI Ethics

3. **Accountability:** Establishing a clear note of accountability is important, especially while operating in critical areas like healthcare or law enforcement. This allows users to learn who would be responsible for the outcomes of AI systems.
4. **Human Dependence:** AI systems can automate many tasks humans previously performed, especially data-driven ones. However, because AI cannot bear moral or legal responsibility, critical decisions, particularly in healthcare, justice, and finance, should not be fully delegated to AI; a human must remain in the loop.
5. **Social Impact:** The use of AI over employment, social interactions, and power dynamics must be carefully considered to promote positive outcomes.

AI Safety & Regulation

1. **Alignment:** Ensuring AI systems pursue the goals their developers and users actually intend, and behave helpfully, honestly, and harmlessly. As models become more capable, misalignment is a growing safety concern.
2. **Regulation:** Governments are introducing AI laws: the EU AI Act (2024) takes a risk-based approach to high-risk systems. The US, UK, and China have issued executive orders and safety frameworks, with many other nations drafting AI strategies.
3. **Catastrophic & Existential Risks:** Researchers warn that powerful AI could be misused for cyberattacks, biosecurity threats, disinformation, or, in the long term, loss of human control. Responsible development requires evaluations, red-teaming, and ongoing safety research.

AI Bias

- ❖ Usually machines shouldn't be biased, as they don't have experiences or memories.
- ❖ AI learns from the data, and if that data is incorrect or misleading, the outputs of the algorithms will also be inaccurate.
- ❖ Some of the common AI biases are:
 1. **Algorithm Bias:** If the algorithm fed to the system itself is faulty, it affects the outcome of the algorithm as well.
 2. **Sample Bias:** If the dataset you picked is irrelevant and not accurate, the error will reflect on the results.
 3. **Prejudice Bias:** Uses data that is influenced by social biases like prejudice and discrimination.

AI Bias

4. **Measurement Bias:** This bias occurs when data is incorrectly collected, measured, and integrated.
5. **Exclusion Bias:** This bias occurs when an important data point is left out of a dataset, often because of human negligence, which could be intentional (not knowing the significance) or by mistake.
6. **Selection Bias:** This bias occurs when the data used to train the algorithm doesn't reflect the real-world distribution.

AI Bias

Prevention of Bias

- ❖ Bias often causes inequality and regulatory challenges.
- ❖ To address these issues, organizations should take certain measures to develop ethical practices.
- ❖ Some of the key measures to prevent bias are:
 1. Most biases occur due to small or limited datasets. To avoid this, collect as much data as possible from multiple sources to diversify the dataset.
 2. Run multiple tests during the early stages of testing to check for biases and correct them.
 3. Continuously check for the data quality as time passes.